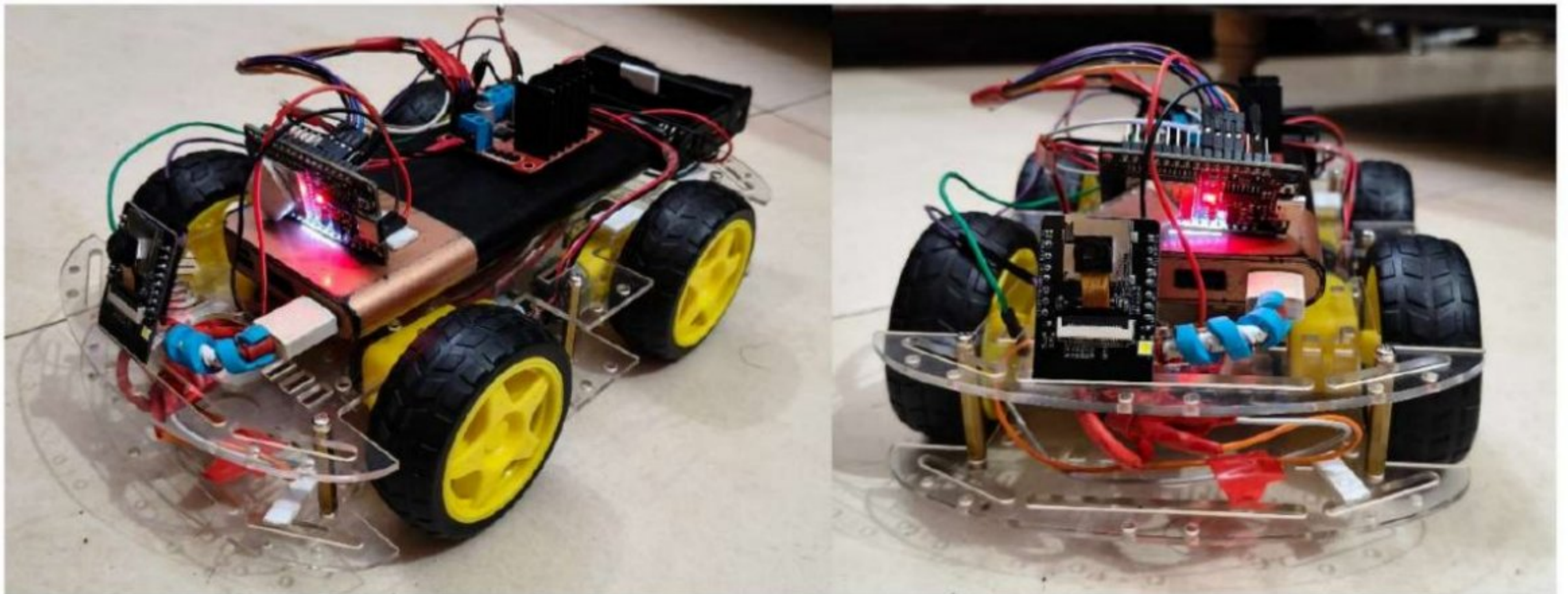


PROJECT REPORT



BY SHUBHAM
ECE student

ESP32-BASED SURVEILLANCE ROVER



This project focuses on designing and building a Wi-Fi-controlled Surveillance Rover equipped with live video streaming capabilities. Using an ESP32 microcontroller, an ESP32-CAM module, and an L298N Motor Driver, the rover can be maneuvered remotely while providing real-time visual feedback over a local wireless network. Power is managed using a dual-supply system (5V Power Bank for microcontrollers and 3 - 18650 Li-ion battery for the motors).

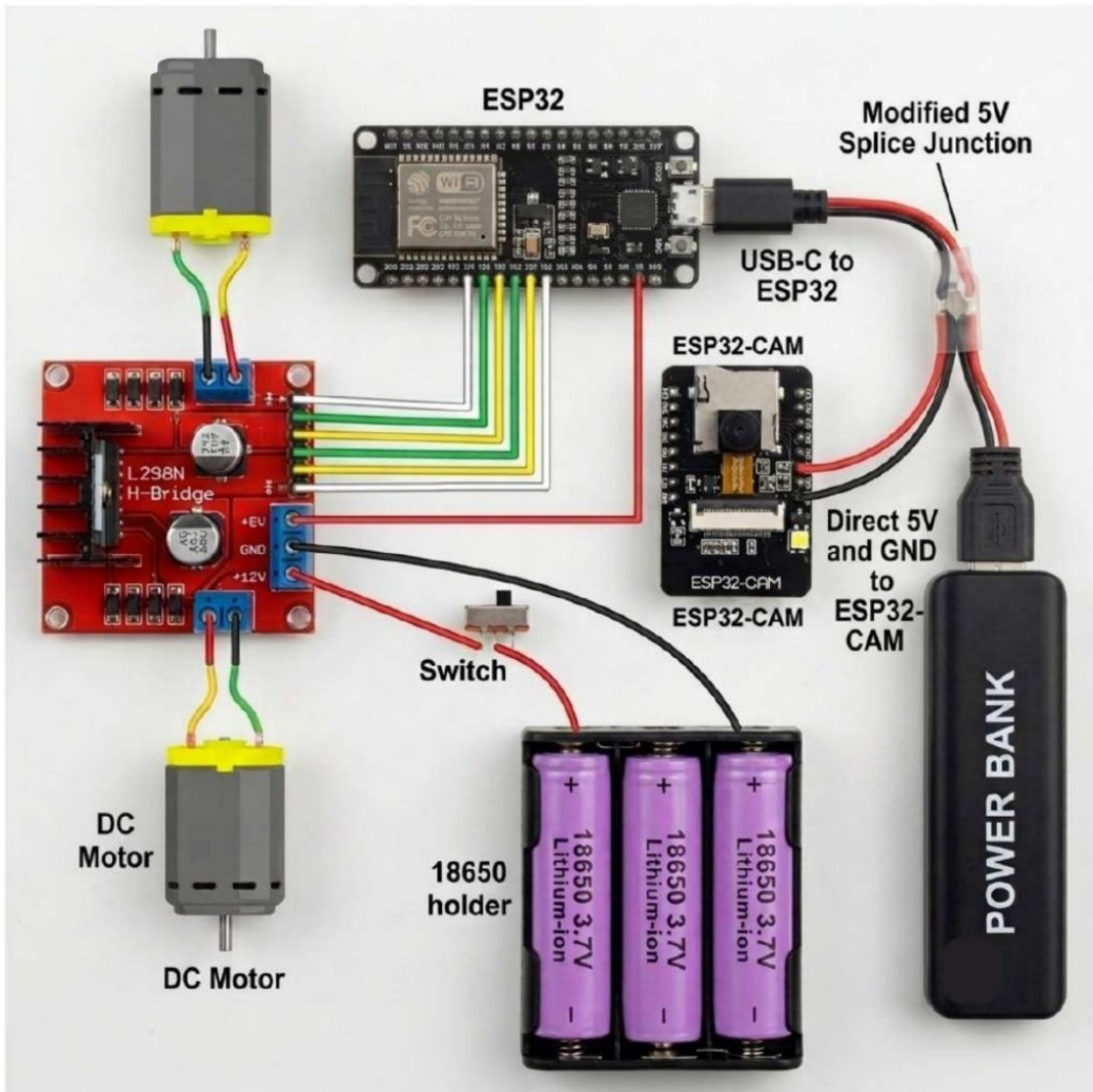
PROJECT OBJECTIVES

- Build a low-cost, compact remote-operated robotic vehicle.
- Implement real-time video streaming via a web-based interface using the ESP32-CAM.
- Establish dual-power management to prevent voltage drops during motor operations.
- Ensure low-latency motor control over Wi-Fi.

KEY FEATURES

- Wireless Control: Real-time motor operation via web interface.
- Live Video Feed: Smooth local camera stream for navigation.
- Dual Power Line: Separate power logic for microcontrollers and high-current drive motors.

CIRCUIT DIAGRAM



COMPONENTS :

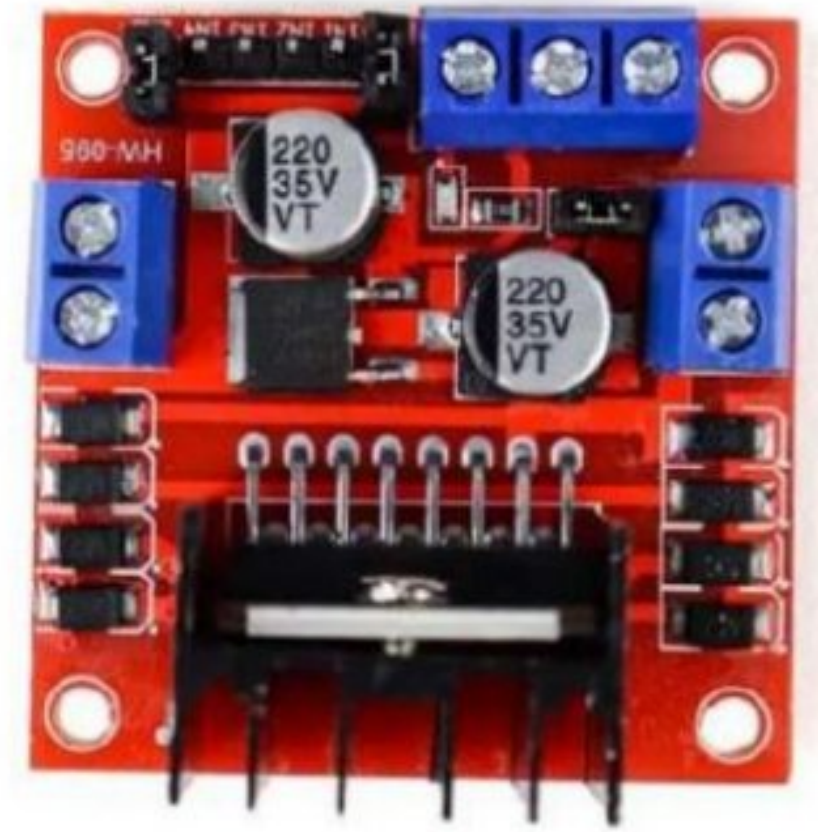
1. ESP 32 MICROCONTROLLER

Acts as the primary control node. It receives directional commands via Wi-Fi and sends PWM signals to the motor driver.



2. L-298 MOTOR DRIVER

Dual H-bridge module that translates low-voltage logic signals from the ESP32 into high-current power for the motors.



3. ESP CAMERA

Equipped with an OV2640 camera sensor to host a local web server for live video streaming.



4. GEARED MOTOR

DC motors fitted with reduction gearboxes to provide the torque necessary for rough terrain navigation.



5. Li batteries

A 5V power bank supplies regulated power to the microcontrollers to avoid brownouts, while a dedicated 18650 battery drives the motors.

